



MR System Engineering
RF-Coil (3T) - Simulation

Protocol

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Project Task

MRI RF Coil Simulation

Background

The core of MR imaging is measuring small signals out of the patient's body. The signal comes from magnetically excited hydrogen protons called spins, that induces a radio-frequency (RF) voltage in a receive (RX) coil. The same coil can also be used to excite the spins by transmitting (TX) a magnetic RF field.

The coil is sensitive to the spin signal by using the resonance effect. For most human 3T systems, the main transmit coil is a volume coil integrated inside the MR system, called a body coil or birdcage. For receiving the tissue signals, local surface coils can be used, consisting of individual loop coils. For higher field strengths, local TX coils can be used as well.

Task

Simulate 2 MRI RF coils of your choice, perform the tuning and matching procedure, and generate B_1^+ , B_1^- and a SAR_{10g} map.

1. Study the basics of an electrical resonator, frequency tuning, and impedance matching. To what frequency do you have to tune the resonator coil?
2. Study MRI coil types, their basic structure, and what they are used for.
3. Study the basics of scattering parameters and how they look for a simple electric resonator. How can you measure the tuning and matching of the resonator coil?
4. Design 2 MRI coils for a frequency of your choice in the CST Studio Suite. Perform the 3D simulation and place the necessary lumped elements on the coils using co-simulation.
5. Tune the coils to your frequency and match them to 50Ω .
6. Calculate B_1^+ , B_1^- , and SAR maps of the two coils and compare them with each other.

Statutory Declaration

We hereby affirm that we have prepared this work independently and with the use of the specified literature and resources only.

Magdeburg, June 24, 2026

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1 Theoretical Background

1.1 Introduction: The RF Coil as a Resonant Circuit

In Magnetic Resonance Imaging (MRI), the Radio Frequency (RF) coil is a critical component responsible for both transmitting the excitation field (B_1^+) and receiving the resulting signal (B_1^-) [1].

At its core, an RF coil is an electrical resonator, specifically designed to oscillate efficiently at a precise frequency. This is achieved by forming an RLC circuit, which consists of a resistor (R), an inductor (L), and a capacitor [2].

The inductor is the coil itself, typically a loop of wire that generates a magnetic field when current flows through it. The capacitor is a discrete electronic component incorporated into the circuit to enable resonance at the desired operating frequency.

In MRI systems, typically, coils are modelled as parallel RLC circuits.

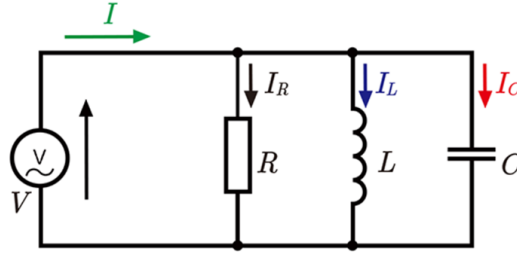


Figure 1: Parallel RLC Circuit [2].

1.2 Larmor Frequency, Tuning, and Impedance Matching

The Larmor equation dictates the operating frequency of an MRI RF coil,

$$f_L = \frac{\gamma}{2\pi} B_0, \quad (1)$$

where $\gamma/(2\pi)$ is the gyromagnetic ratio of the nucleus under investigation. For hydrogen protons (^1H), $\gamma/(2\pi) = 42.58 \text{ MHz/T}$ [1]. In an ideal 3 T magnet, this yields a Larmor frequency of $f_L = 127.74 \text{ MHz}$.

In practice, the actual frequency may differ slightly due to magnet tolerances and loading conditions; the 3 T system used in this work operates at 123 MHz. All CST simulations were therefore performed with a target frequency of 123 MHz.

1.2.1 Tuning and Matching in a Parallel Resonant Circuit

An MRI surface coil behaves as a parallel RLC resonator. Its resonant frequency is given by:

$$f_0 = \frac{1}{2\pi\sqrt{LC}}. \quad (2)$$

Tuning is the process of setting the resonant frequency exactly to the Larmor frequency (123 MHz in this work). This is accomplished by adjusting a variable tuning capacitor C_t . Changing C_t alters the total capacitance of the circuit and therefore shifts the resonance according to Eq. (2).

Matching is the process of transforming the coil impedance to the standard 50Ω system impedance. This is achieved with a matching capacitor C_m . Adjusting C_m changes the impedance seen by the transmit/receive system, minimising the reflected power. Without proper matching, a large fraction of the RF power would be reflected back to the amplifier, reducing efficiency and potentially causing damage. The quality of both tuning and matching is evaluated by the reflection coefficient S_{11} .

1.3 MRI RF Coil Types: Volume, Surface, and Dipole Antennas

MRI RF coils are traditionally divided into volume coils and surface coils, with dipole antennas becoming increasingly important at high fields.

Volume coils completely enclose the anatomy and are designed to generate a highly homogeneous transverse magnetic field. The most common design is the birdcage coil, consisting of two circular end-rings connected by equally spaced straight rungs. Capacitors placed on the end-rings allow tuning to the Larmor frequency. By feeding two ports in quadrature (90° phase shift), the birdcage produces a circularly polarised B_1^+ field, which reduces the required transmit power by half compared with a linearly polarised coil [3].

Surface coils are placed directly on or near the region of interest. The simplest form is a single loop of copper conductor, with distributed capacitors to create the required capacitance. Surface coils provide a very high signal-to-noise ratio (SNR) immediately adjacent to the coil, but their sensitivity falls off rapidly with distance. They are therefore predominantly used as receive-only elements [1].

Dipole antennas have been introduced to overcome the challenges that appear at 3 T and above, notably the shortened wavelength and increased SAR [4]. A common design is the meandered dipole, where the conductor is folded to increase the effective inductance and reduce the physical length, while also lowering mutual coupling between array elements.

1.4 Scattering Parameters and the S_{11} of a Parallel Resonator

Scattering parameters (S-parameters) relate incident and reflected voltage waves at the ports of a high-frequency network [5]. For a one-port device such as an RF coil, only the input reflection coefficient S_{11} is needed:

$$S_{11} = \frac{Z_{in} - Z_0}{Z_{in} + Z_0}, \quad (3)$$

where Z_{in} is the complex input impedance after the matching network and $Z_0 = 50 \Omega$. In decibels,

$$|S_{11}|_{dB} = 20 \log_{10} \left| \frac{Z_{in} - 50}{Z_{in} + 50} \right|. \quad (4)$$

A perfect match ($Z_{in} = 50 \Omega$) gives $S_{11} \rightarrow -\infty$ dB; a complete mismatch ($Z_{in} = 0$ or ∞) corresponds to 0 dB. An $|S_{11}|$ value below -30 dB (less than 0.1 % reflected power) is considered excellent for MRI applications.

A parallel RLC resonator without any matching network presents an impedance at resonance equal to R_p (several k Ω) [2]. Because $R_p \gg 50 \Omega$, the native $|S_{11}|$ is close to 0 dB over a broad frequency range. Introducing the tuning capacitor C_t and matching capacitor C_m transforms this high impedance to 50Ω at the target frequency. The resulting $|S_{11}|$ curve then exhibits a sharp dip centred at f_L ; the position of the dip indicates the tuning quality, and its depth reflects the matching quality [5].

1.5 Measurement of Tuning and Matching

1.5.1 Measurement with a Vector Network Analyser

The standard procedure for verifying tuning and matching is performed with a vector network analyser (VNA) [6]:

1. **Calibration:** The VNA, together with the coaxial cable, is calibrated using an Open–Short–Load (OSL) kit to shift the reference plane to the coil input.
2. **Loading:** The coil is placed on the tissue-equivalent phantom to reproduce the electromagnetic load.
3. **Frequency sweep:** The VNA sweeps a frequency range that covers the target Larmor frequency and displays $|S_{11}|$.

4. **Tuning:** The tuning capacitor C_t is varied until the minimum of the $|S_{11}|$ curve is centred at 123 MHz.
5. **Matching:** The matching capacitor C_m is then adjusted to deepen the dip as much as possible.
6. **Iteration:** Because tuning and matching are interdependent, steps 4 and 5 are repeated iteratively until the dip remains centred at 123 MHz and its magnitude is below -30 dB.

1.5.2 Simulation-Based Tuning and Matching in CST Co-Simulation

In CST Studio Suite, the same iterative procedure is carried out virtually using the co-simulation environment. After the 3D EM simulation, the model is imported into the schematic, where the discrete capacitor ports are connected to lumped capacitors whose values are parameterised (e.g., C_t and C_m). An S-parameter simulation task is then defined over a frequency range around 123 MHz.

The initial capacitor values are set to suggested starting points (e.g., $C_t = 30$ pF, $C_m = 3$ pF) and the S-parameter sweep is executed. The resulting $|S_{11}|$ curve is inspected:

- If the dip is not centred at 123 MHz, C_t is adjusted.
- If the dip is centred but not deep enough, C_m is adjusted.

These modifications are repeated over several iterations until the $|S_{11}|$ minimum occurs exactly at 123 MHz and reaches a value below -30 dB. Only then is the coil considered properly tuned and matched in the simulation, and the corresponding field distributions are extracted for B_1^+ , B_1^- , and SAR analysis.

2 Materials and Methods

2.1 Modeling of the Circular Surface Loop Coil and Substrate

The geometry of the circular surface loop coil was created in **CST Studio Suite**.

The coil conductor was realised as a two-dimensional (2D) sheet of Perfect Electric Conductor (PEC), which accurately represents the 35 μm copper layer while keeping the mesh efficient. The loop is circular and contains four gaps equally spaced by 90° along its circumference.

A circular **substrate** made of FR-4 (1.5 mm thick) supports the coil and extends slightly beyond its outer edge.

Four **discrete ports** ($50\ \Omega$ reference impedance each) were placed at the four gap positions. The detailed port assignment and the connection of tuning/matching capacitors are described in the following subsections.

Figure 2 shows the CST model with all dimensions and material assignments.

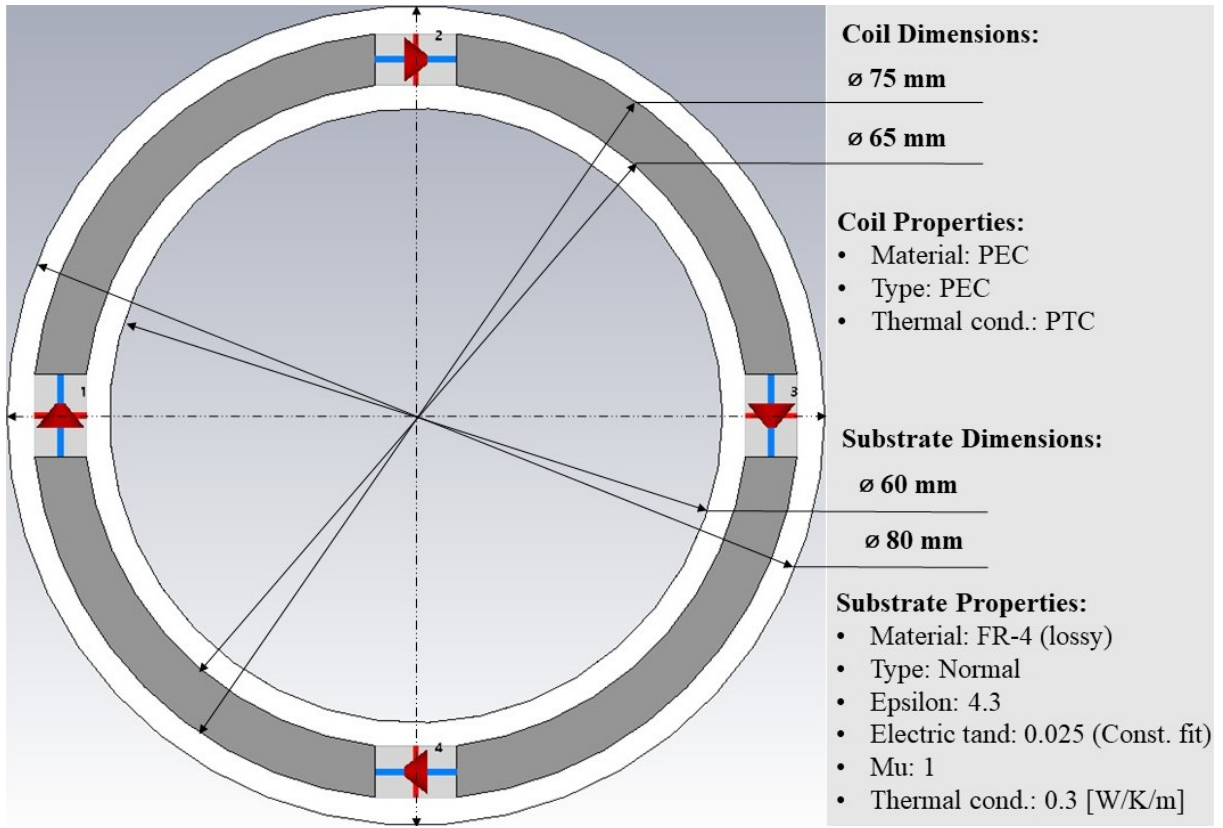


Figure 2: Surface loop coil(indicated in gray), FR-4 substrate (indicated in white), and the four discrete ports equally spaced by 90° .

2.2 Phantom Modeling and Coil–Phantom Distance

A **tissue-equivalent phantom** was modelled to provide realistic electromagnetic loading of the coil. Following the specifications of the Siemens Healthineers cylindrical load phantom, the phantom was realised as a **homogeneous cylinder** filled with a material(water) that mimics the dielectric properties of human tissue at radio frequencies.

The cylinder axis was oriented along the **z -direction** of the coordinate system, which is required for the correct calculation of B_1^+ and B_1^- in post-processing. The exact diameter and length of the phantom are indicated in Figure 3.

The coil was positioned such that the distance between the **conducting layer (PEC)** and the top of the phantom is exactly **10 mm**, as prescribed by the design guidelines. The FR-4 substrate (thickness

1.5 mm) is located between the conductor and the phantom, with its bottom face facing the phantom. Consequently, the physical gap between the substrate surface and the phantom is 8.5 mm, but the effective coil-to-phantom distance remains the required 10 mm.

This configuration ensures strong inductive coupling while leaving the conductor accessible from above for tuning and matching adjustments, exactly as in a practical laboratory setup.

Figure 3 shows a side view of the coil–phantom assembly, with all relevant dimensions, material types, and the direction of the z -axis indicated.

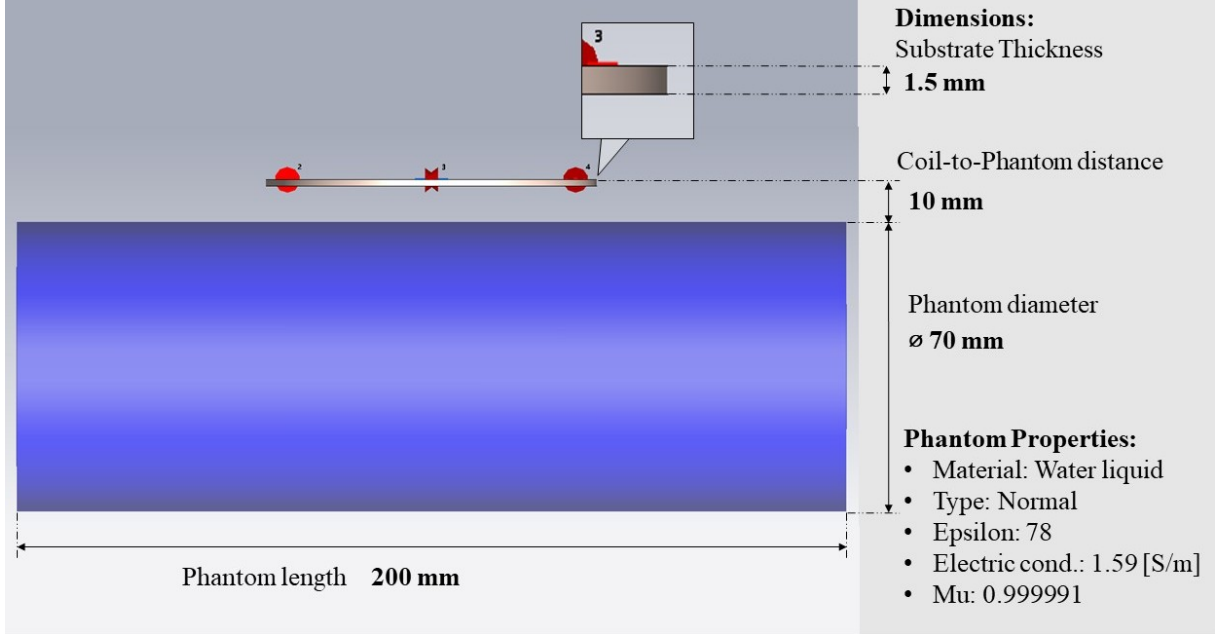


Figure 3: Side view of the coil and phantom: the PEC conductor is 10 mm above the phantom, with the FR-4 substrate (1.5 mm) in between. The substrate faces the phantom.

2.3 Circuit Schematic and Lumped Element Connections

After the 3D electromagnetic simulation, the four discrete ports that represent the coil gaps were transferred into the CST Design Studio co-simulation environment.

All ports were converted to differential operation, i.e. each port is treated as a terminal pair: Port 1 as (1, 1'), Port 2 as (2, 2'), Port 3 as (3, 3'), and Port 4 as (4, 4'). The ports are equally spaced by 90° around the loop, consistent with the layout of the 3D model.

The lumped capacitors were assigned as follows:

- **Ports 2, 3, and 4** – each receives a **tuning capacitor** C_t . These three identical capacitors determine the resonant frequency of the parallel-RLC coil.
- **Port 1** – the feed point. Here, a **matching capacitor** C_m is placed in **series** with the loop, while a **tuning capacitor** C_t is connected in **parallel**.

This network simultaneously adjusts the input impedance to 50Ω and contributes to the overall tuning of the coil.

An external 50Ω excitation port was connected to the junction of C_m and C_t to represent the coaxial feed of the MRI system.

Figure 4 shows the actual implementation of this circuit in the CST Design Studio schematic interface.

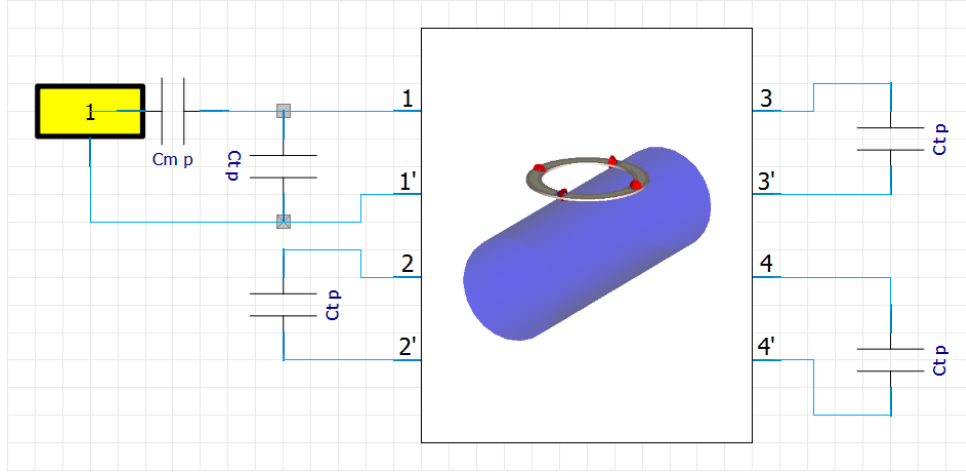


Figure 4: Co-simulation schematic in CST Design Studio, showing the four differential ports, the distributed capacitors C_t on ports 2–4, and the feed network with series C_m and parallel C_t on port 1.

2.4 Mesh Settings

Accurate electromagnetic simulation of the coil and phantom requires a mesh that resolves both the geometric details of the conductors and the wavelength inside the high-permittivity materials. The time-domain solver uses a hexahedral mesh based on the Finite Integration Technique (FIT). In this method, the computational volume is divided into rectangular cuboids of variable size, and the electric and magnetic fields are updated alternately in time. The mesh cell size is constrained by two conditions. First, the smallest cell determines the maximum allowed time step through the Courant stability criterion, so it must be kept as large as possible while still resolving the finest geometric details. Second, the largest cell must sample the shortest wavelength of interest with at least 10 cells per wavelength to avoid numerical dispersion and ensure accurate field propagation. Therefore, local mesh properties were assigned individually to the three critical components: the coil conductor, the substrate, and the phantom.

For the **coil conductor**, a maximum cell size of **2 mm** was prescribed in all three spatial directions. Since the conductor width is 5 mm, this guarantees at least two to three mesh cells across the strip, allowing the solver to capture the current distribution and the magnetic field with sufficient precision.

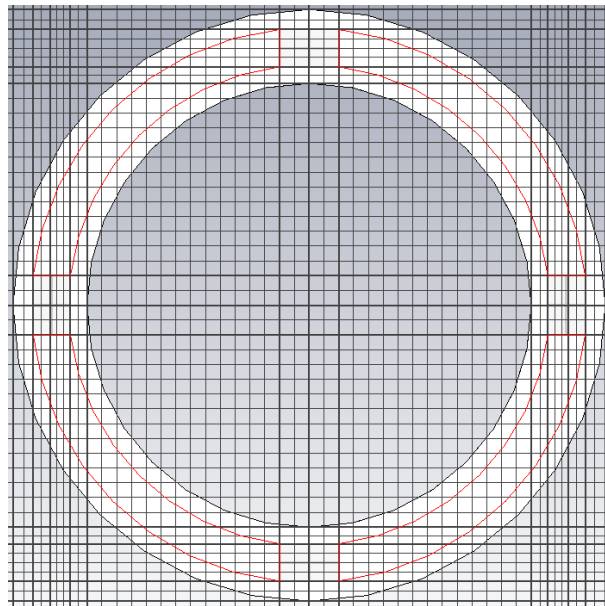


Figure 5: Local mesh view of the coil conductor. The conductor edges are highlighted by red lines.

The **FR-4 substrate**, which is only 1.5 mm thick, was meshed using the number of cells across object criterion set to **4**.

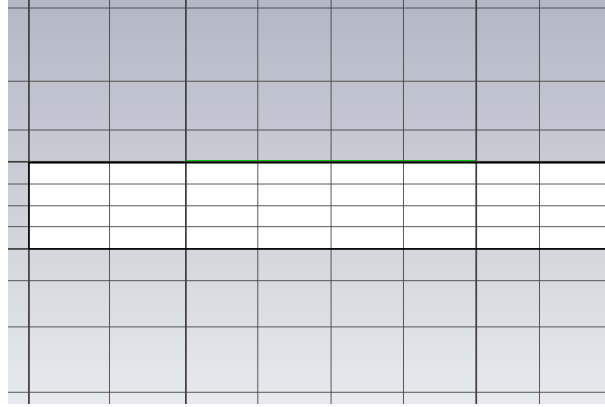


Figure 6: Local mesh view of the FR-4 substrate.

For the **cylindrical phantom**, an absolute cell size of **2 mm** was chosen.

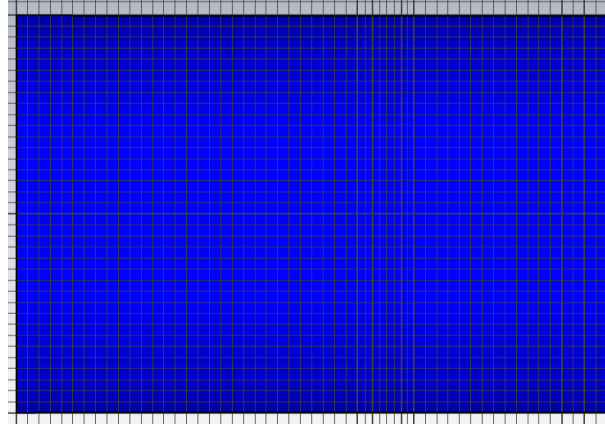


Figure 7: Local mesh view of the cylindrical phantom, shown as a half-cylinder for visualisation.

Such fine resolution is recommended for reliable SAR and near-field calculations, as it minimises numerical dispersion and allows the local specific absorption rate to be evaluated with high spatial accuracy.

2.5 Iterative Tuning and Matching in Co-Simulation

Following the electromagnetic simulation of the 3D model, the co-simulation environment of CST Design Studio was used to determine the capacitor values that place the resonance at 123 MHz and match the coil to 50Ω . The initial values were set to $C_t = 30\text{ pF}$ and $C_m = 3\text{ pF}$, as recommended in the simulation guidelines.

The tuning and matching process was carried out as a manual iterative optimisation. In each step, the $|S_{11}|$ curve was inspected, C_t was adjusted to shift the resonance frequency to 123 MHz, and C_m was then modified to deepen the reflection minimum. These two adjustments were repeated until the dip of $|S_{11}|$ was centred at the target frequency, and its magnitude reached a satisfactory level.

Approximately **50 iterations** were required to converge to the final values:

- $C_t = 42.25\text{ pF}$,
- $C_m = 10.0\text{ pF}$.

The resulting $|S_{11}|$ curve is shown in Figure 8. At 123 MHz the reflection coefficient is around -34.27 dB, well below the -30 dB threshold considered excellent in MRI. The fraction of reflected power can be calculated directly from the definition of the reflection coefficient in decibels. Rearranging Eq. (4),

$$\frac{P_{\text{reflected}}}{P_{\text{incident}}} = |S_{11}|^2 = \left(10^{|S_{11}|_{\text{dB}}/20}\right)^2.$$

Substituting the measured value,

$$|S_{11}| = 10^{-34.27/20} \approx 0.0194,$$

$$|S_{11}|^2 \approx (0.0194)^2 \approx 3.77 \times 10^{-4} \approx 0.038 \%.$$

Thus, approximately **0.04 %** of the incident power is reflected, confirming that the coil is both accurately tuned and efficiently matched.

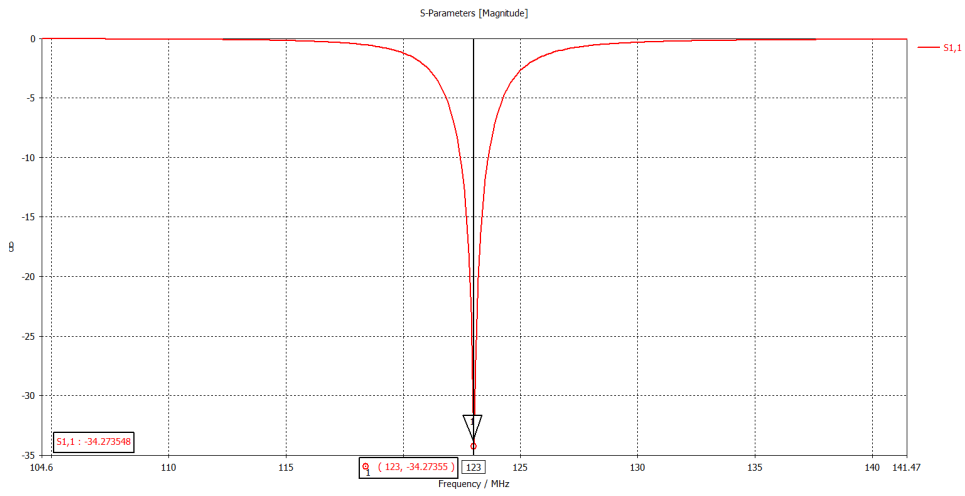


Figure 8: Magnitude of S_{11} after iterative tuning and matching. The minimum is located at 123 MHz with a value of -34.27 dB.

3 Results

3.1 Transmit Field B_1^+ Distribution

Figures 9–11 show the B_1^+ magnitude maps in three orthogonal centre planes and at the phantom surface.

The B_1^+ field is strongest immediately beneath the coil at $y = 0$, reaching a peak of approximately $12\text{--}13\text{ }\mu\text{T}$ inside the phantom near its upper surface. The field decays rapidly with increasing depth into the phantom, which is the characteristic behaviour of a surface coil providing high localised SNR near the coil but limited penetration depth. In the XY and XZ planes the field is spatially confined to the region beneath the coil aperture ($|x| \leq 37\text{ mm}$, $|z| \leq 37\text{ mm}$) and vanishes toward the phantom ends ($|z| > 100\text{ mm}$). Figure 11 shows the B_1^+ distribution at $y = -10\text{ mm}$, i.e. at the top surface of the phantom, where the RF field enters the tissue.

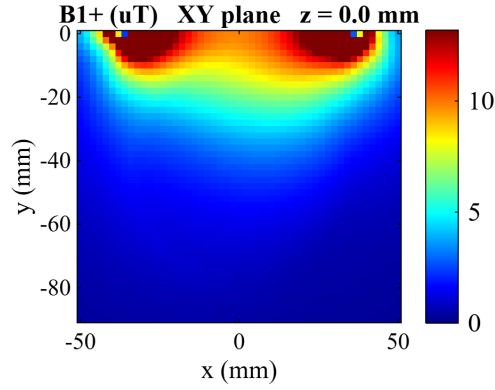


Figure 9: B_1^+ magnitude map in the XY plane at $z = 0\text{ mm}$ (centre of the phantom along the bore axis). Colour scale: $0\text{--}13\text{ }\mu\text{T}$.

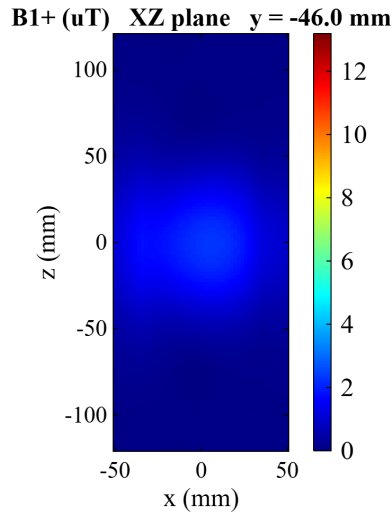


Figure 10: B_1^+ magnitude map in the XZ plane at $y = -46\text{ mm}$ (coil plane). The field is beneath the coil aperture and decays toward the phantom edges.

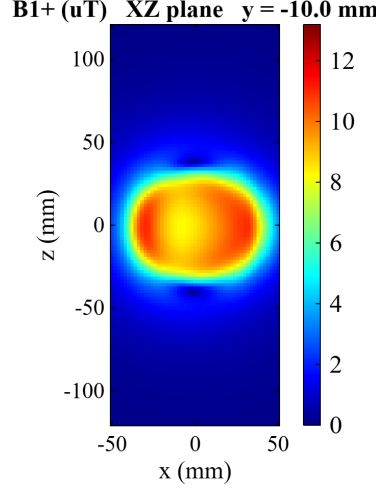


Figure 11: B_1^+ magnitude map in the XZ plane at $y = -10$ mm, corresponding to the top surface of the phantom. This slice represents the entry point of the RF field into the tissue.

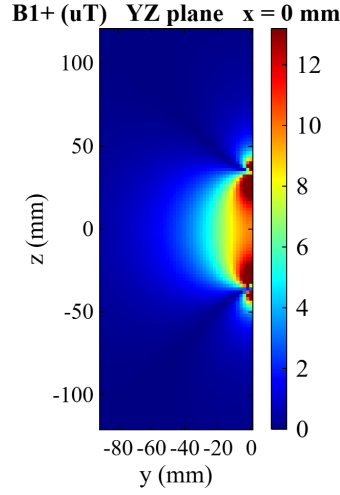


Figure 12: B_1^+ magnitude map in the YZ plane at $x = 0$ mm. The rapid depth-wise decay of the transmit field with increasing distance from the coil is clearly visible.

3.2 Receive Field B_1^- Distribution

The receive sensitivity B_1^- was computed from the same co-simulation results following the reciprocity principle. Figures 13–15 present the corresponding magnitude maps.

The B_1^- distribution is nearly identical to the B_1^+ distribution in both spatial pattern and peak magnitude (12–13 μ T). This result is expected for a single linear loop coil, which transmits and receives the same linearly polarised field component. The close agreement between B_1^+ and B_1^- confirms the correct implementation of the reciprocity-based post-processing in CST.

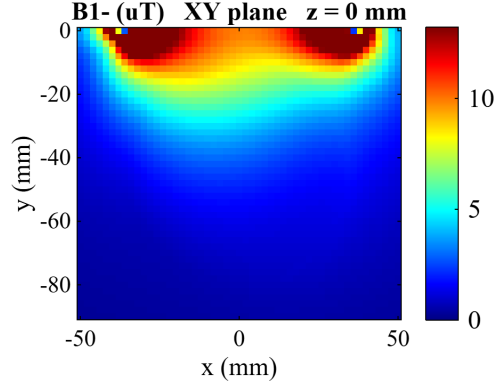


Figure 13: B_1^- magnitude map in the XY plane at $z = 0$ mm. The receive sensitivity pattern closely mirrors the B_1^+ distribution shown in Figure 9.

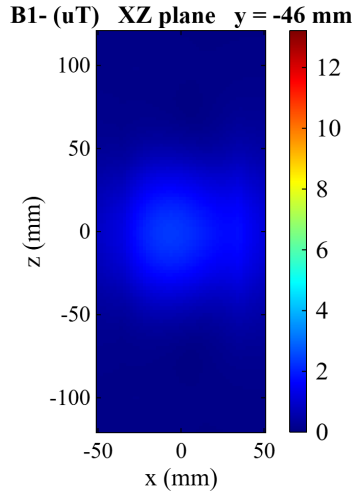


Figure 14: B_1^- magnitude map in the XZ plane at $y = -46$ mm (coil plane).

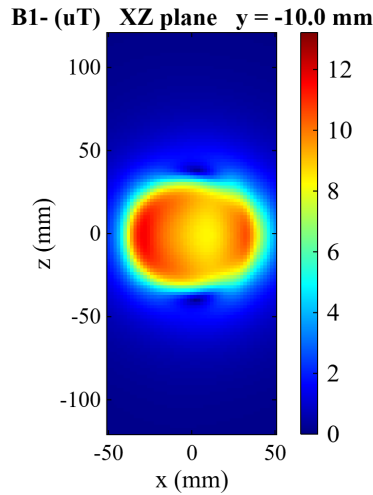


Figure 15: B_1^- magnitude map in the XZ plane at $y = -10$ mm (phantom top surface). The pattern is near-identical to the corresponding B_1^+ slice in Figure 11, as expected for a linear loop coil.

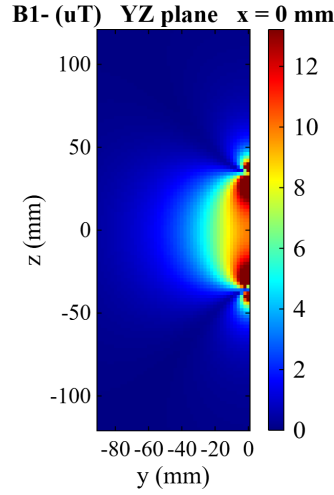


Figure 16: B_1^- magnitude map in the YZ plane at $x = 0$ mm.

3.3 Specific Absorption Rate (SAR_{10g}) Distribution

The 10-g averaged SAR was calculated from the co-simulation power loss density, normalised to 1 W stimulated power. Figures 17–19 present the SAR maps in three orthogonal planes.

The SAR_{10g} map in the XY plane at $z = 0$ mm (Figure 17) shows the highest power deposition concentrated at the upper surface of the phantom, directly beneath the coil. The SAR decreases with increasing depth into the phantom, following the expected near-field decay of a surface loop coil.

The SAR_{10g} map in the XZ plane at $y = -46$ mm (Figure 18) shows two parallel bright bands running along the z -direction. This slice cuts longitudinally through the cylindrical phantom, and the two bands represent the left and right outer walls of the cylinder at $x = \pm 35$ mm, visible as two parallel edges in the cross-section — analogous to cutting a hollow cylinder along its length and viewing the two resulting edges face-on. Outside the phantom volume ($|x| > 35$ mm) and beyond the phantom ends ($|z| > 100$ mm) the SAR drops to zero, confirming correct spatial confinement of the RF heating.

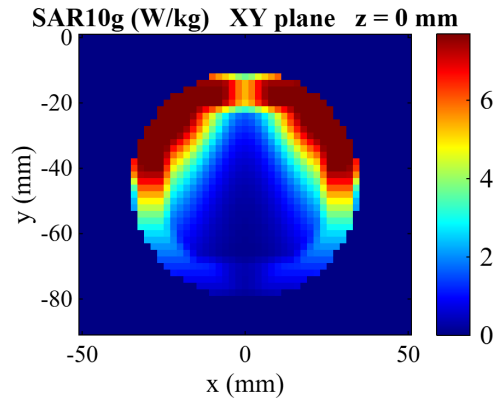


Figure 17: 10-g averaged SAR map in the XY plane at $z = 0$ mm. The highest power deposition is concentrated at the phantom surface directly beneath the coil. Colour scale: 0–8 W/kg.

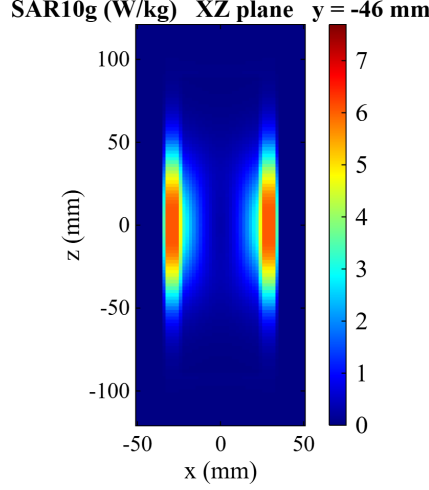


Figure 18: 10-g averaged SAR map in the XZ plane at $y = -46$ mm. Two parallel bands at $x = \pm 35$ mm correspond to the left and right outer walls of the cylindrical phantom at this longitudinal cross-section.

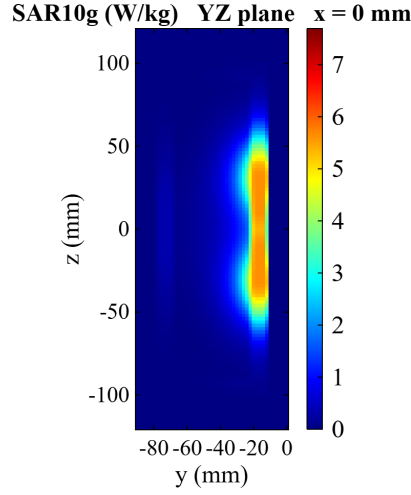


Figure 19: 10-g averaged SAR map in the YZ plane at $x = 0$ mm, showing the depth penetration profile of the SAR into the phantom along the y -direction.

3.4 Surface Current Distribution

Figure 20 shows the three-dimensional distribution of the surface current density $|J_S|$ on the PEC conductor of the coil at 123 MHz. Figure 21 presents the corresponding top-down view in the XZ plane.

The peak surface current density reached approximately $|J_S|_{\max} \approx 95$ A/m, with the highest values concentrated near the port and capacitor gap locations at 0° , 90° , 180° , and 270° . Between the gaps the distribution is relatively uniform, consistent with a well-tuned distributed-capacitor design. Small deviations near the ports are expected because the lumped elements locally perturb the current path.

Surface Current Density $|J_S|$ on RF Coil Conductor
123 MHz | 1W stimulated power | peak = 95.025 A/m

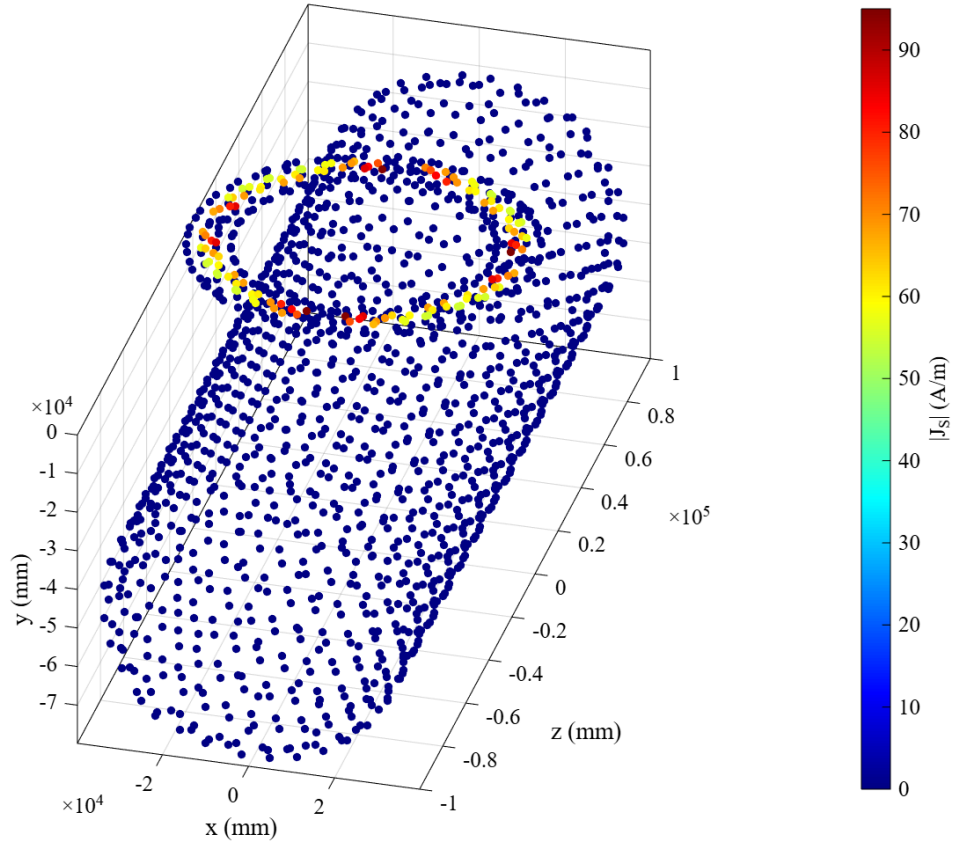


Figure 20: Three-dimensional scatter plot of the surface current density $|J_S|$ (A/m) on the RF coil conductor at 123 MHz. Red indicates peak current; blue indicates minimum current. Peak = 95.025 A/m.

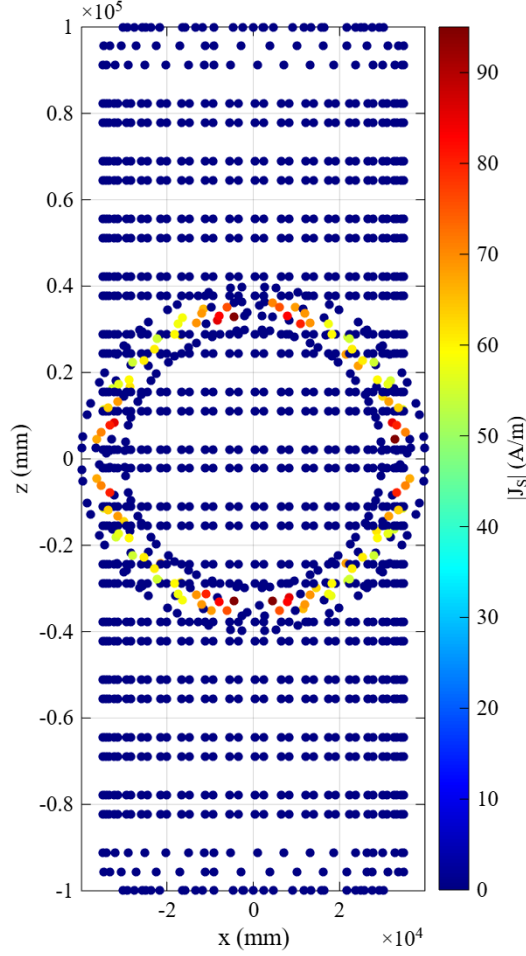


Figure 21: Top-down view (XZ plane) of the surface current density $|J_s|$ on the coil ring. Peak values are visible near the four port and capacitor gap positions at 0° , 90° , 180° , and 270° .

3.5 Power Budget and Loss Distribution

The power budget of the tuned coil at 123 MHz was examined using the 1D results from the CST Design Studio co-simulation. Table 1 summarises the recorded values.

Table 1: Power budget of the circular surface loop at 123 MHz. All values are fractions of the incident power (1 W stimulated).

Component	Fraction of incident power
Accepted power	0.99959222
Outgoing power (all ports)	0.00054125506
Volume loss in phantom liquid	0.9929033
Volume loss in FR-4 substrate	0.0006433525
Loss in dielectrics (total)	0.99355
Sum (Accepted + Outgoing)	1.00013347506

The accepted power reaches 0.99959222, confirming that over 99.95 % of the incident power enters the coil–phantom system. The outgoing power is only 0.00054125506, consistent with the reflection coefficient $|S_{11}| \approx -34.3$ dB.

Nearly all of the accepted power is dissipated in the phantom liquid (0.9929033, i.e. 99.29 %), which is the intended physical mechanism for B_1^+ generation and SAR. The loss in the FR-4 substrate is negligible (0.0006433525, i.e. 0.06 %).

In percentage terms, the power budget is summarised as follows:

- **99.29 %** — dissipated in the phantom (useful transmit power),
- **0.06 %** — lost in the FR-4 substrate,
- **0.60 %** — radiated and numerical residuals,
- **0.054 %** — reflected back to the source.

The sum of accepted and outgoing power deviates from unity by only 0.013 %, which is within the numerical tolerance of the time-domain solver. The power budget therefore confirms that the coil is excellently matched and delivers virtually all available power into the phantom.

3.6 Summary of Key Simulation Results

Table 2 summarises the principal quantitative results obtained from the simulation of the 3 T circular surface loop coil.

Table 2: Summary of simulation results for the 3 T circular surface loop coil at 123 MHz.

Quantity	Value	Location
$ S_{11} $ minimum	−34.27 dB	$f = 123$ MHz
Tuning capacitor C_t	42.25 pF	Ports 1–4
Matching capacitor C_m	10.0 pF	Port 1 (series)
Outgoing power	0.054 %	Ports 1–4
B_1^+ peak*	$\approx 12\text{--}13\ \mu\text{T}$	Coil plane, $y = 0$ mm
B_1^- peak*	$\approx 12\text{--}13\ \mu\text{T}$	Coil plane, $y = 0$ mm
$\text{SAR}_{10\text{g}} \text{ max}^*$	8.26 W/kg	Phantom surface beneath coil
$ J_S $ peak	95.03 A/m	Near capacitor gaps

*Normalised to 1 W stimulated power.

4 Discussion

4.1 Key Findings

This work demonstrated the complete simulation workflow for a 3 T MRI circular surface loop coil, from geometry modelling in CST Studio Suite through to the extraction of B_1^+ , B_1^- , and SAR_{10g} field maps. The iterative co-simulation tuning procedure converged to $C_t = 42.25$ pF and $C_m = 10.0$ pF, yielding an $|S_{11}|$ of -34.27 dB at 123 MHz. This value is well below the -30 dB threshold considered excellent for MRI applications. The total outgoing power from all ports was 0.054 %. The sharp, well-centred resonance dip demonstrates that the distributed-capacitor topology, with three tuning capacitors at Ports 2,3,4 and one series matching capacitor with one parallel tuning capacitor at the Feed Port, is an effective strategy for simultaneously achieving resonance and impedance transformation to $50\ \Omega$.

The B_1^+ and B_1^- magnitude maps were found to be nearly identical in both spatial distribution and peak amplitude ($12\text{--}13\ \mu\text{T}$ at the coil plane). This result is physically expected for a single linear loop coil: the same linearly polarised field component is used for both transmission and reception, and the reciprocity principle therefore predicts equal transmit and receive sensitivity patterns. The rapid decay of both fields with increasing depth into the phantom confirms the well-known characteristic of surface coils, namely high localised signal-to-noise ratio (SNR) in tissue directly beneath the coil, but limited penetration depth compared with volume coils such as the birdcage.

The depth at which the B_1^+ field remains sufficiently strong determines the usable imaging depth of the coil. The results show that the field is concentrated within approximately the upper 30–40 mm of the phantom beneath the coil, which defines the effective field of view for image acquisition. This makes the simulated coil suitable for shallow imaging applications such as skin and superficial muscles where high localised SNR close to the surface is more important than deep tissue coverage.

The SAR_{10g} map in the XY plane at $z = 0$ mm (Figure 17) shows the highest power deposition concentrated at the upper surface of the phantom, directly beneath the coil. The SAR decreases with increasing depth into the phantom, following the expected near-field decay of a surface loop coil. The SAR_{10g} map in the XZ plane at $y = -46$ mm (Figure 18) shows two parallel absorption bands running along the z -direction at $x = \pm 35$ mm. These correspond to the left and right curved surface edges of the cylindrical phantom (radius 35 mm).

4.2 Limitations of the Present Work

A limitation of this work is the use of a homogeneous cylindrical water phantom rather than an anatomically realistic human model. While the water phantom provides a reproducible and well-characterised electromagnetic load, it does not capture the dielectric heterogeneity of real tissue, in which organs, bones, fat, and muscle each have distinct permittivity and conductivity values. The SAR distribution in a real subject would therefore differ from the results presented here.

Furthermore, the simulated $|S_{11}|$ result was not validated against a physical bench measurement using a vector network analyser (VNA). Without this comparison, it is not possible to confirm that the simulated capacitor values would produce the same resonance behaviour in a real fabricated coil, where parasitic effects, soldering joints, and connector impedances may shift the resonance frequency.

The conductor was modelled as a Perfect Electric Conductor (PEC), and therefore ohmic losses in the copper trace were not considered in the simulation. In a real coil, the finite conductivity of copper would introduce additional power loss in the conductor itself, slightly reducing the quality factor and the accepted power delivered to the phantom. This effect is typically small for surface coils at 3 T but may become noticeable in high-field applications or for very thin conductor layers.

Finally, while the mesh settings used (2 mm maximum cell size for both the conductor and the phantom) are sufficient for reliable near-field and SAR calculations, further refinement near the conductor edges could improve the accuracy of the surface current distribution and the fringing field at the capacitor gaps.

4.3 Possible Improvements

The phantom used in this work was a generic cylinder of 70 mm diameter and 200 mm length. A more practical approach would be to use a phantom representative of the anatomical region of interest, for example a spherical phantom of approximately 170 mm diameter for the human head, or a larger elliptical phantom for the chest or abdomen, which would provide a more realistic assessment of field penetration and SAR distribution.

CST Studio Suite was chosen for this project because it provides a convenient integrated co-simulation environment with a built-in MRI Toolbox. However, it requires an expensive commercial licence and does not include anatomical models. Furthermore, software instability was observed during the workflow, where unexpected errors required the application to be fully closed and reopened before work could continue, adding unnecessary time to the project. In future work, cross-validation using one of the aforementioned tools would strengthen confidence in the simulation results.

Several alternative simulation tools exist for MRI coil design, each offering specific advantages that could address the limitations encountered with CST Studio Suite.

Sim4Life (ZMT Zurich) is specifically designed for MRI safety assessment and includes validated anatomical human models, which would overcome the limitation of using a homogeneous phantom.

ANSYS HFSS provides higher geometric accuracy for curved structures through its FEM solver with adaptive tetrahedral meshing, avoiding the staircase approximation of hexahedral grids.

openEMS offers a free, open-source FDTD alternative suitable for academic use without a commercial licence, though it lacks a graphical user interface and an integrated tuning environment.

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Appendices

A Transmit Field B_1^+ – Perspective View

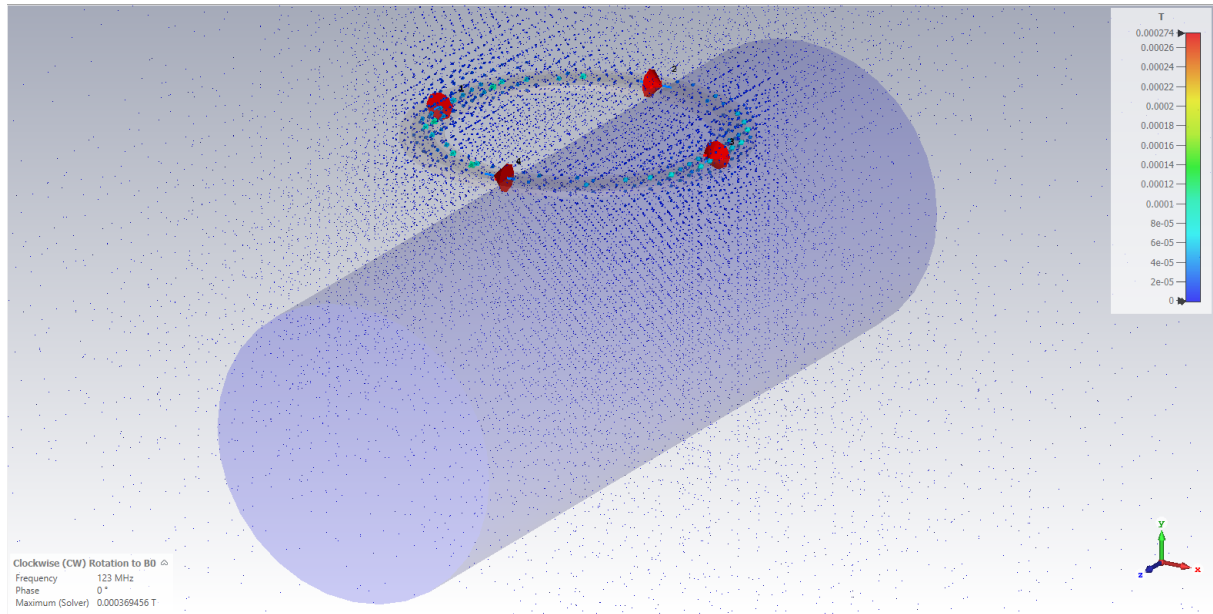


Figure 22: Three-dimensional rendering of the B_1^+ distribution.

B Receive Field B_1^- – Perspective View

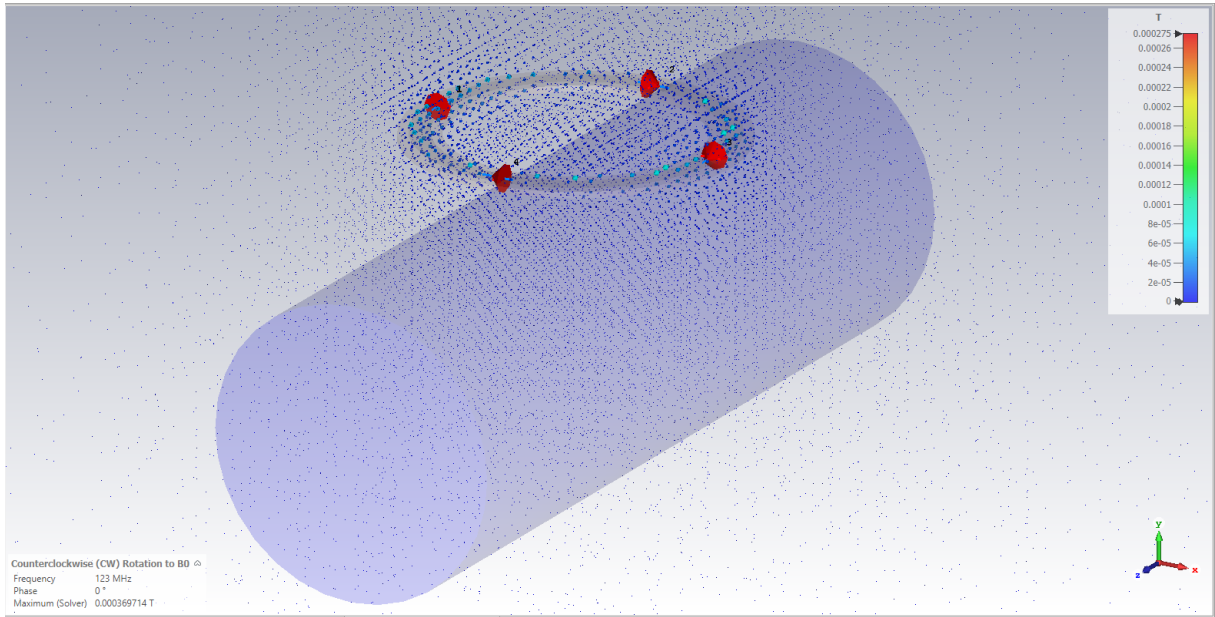


Figure 23: Three-dimensional rendering of the B_1^- distribution.

C SAR (10-g Average) – Perspective View

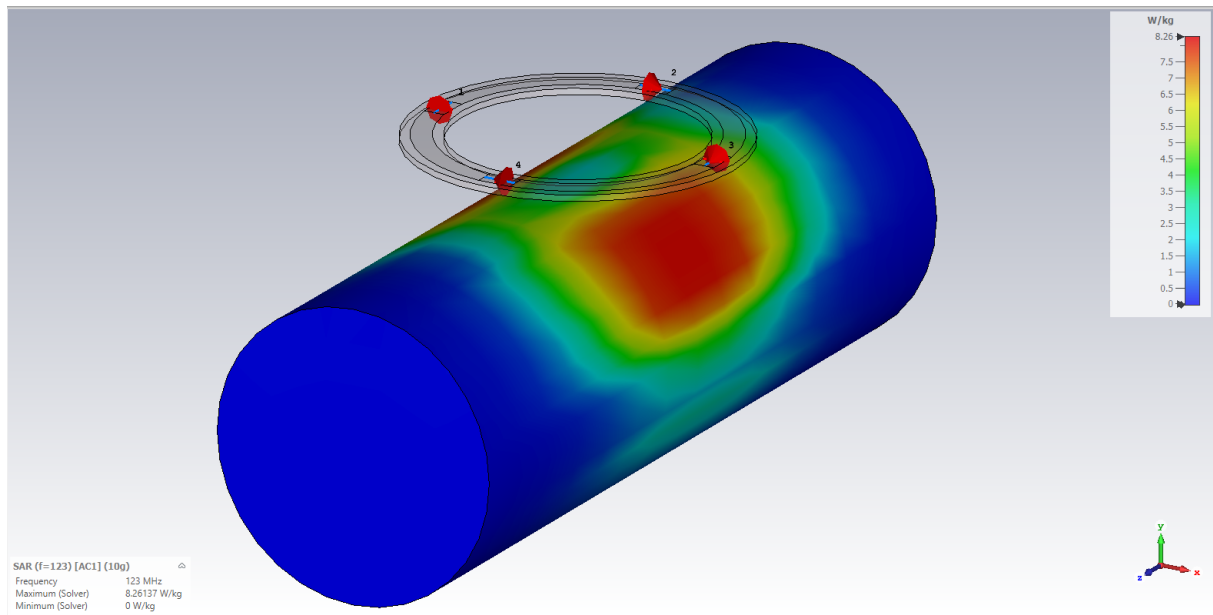


Figure 24: Three-dimensional rendering of the local 10-g averaged SAR.

D Accepted Power – 1D Plot

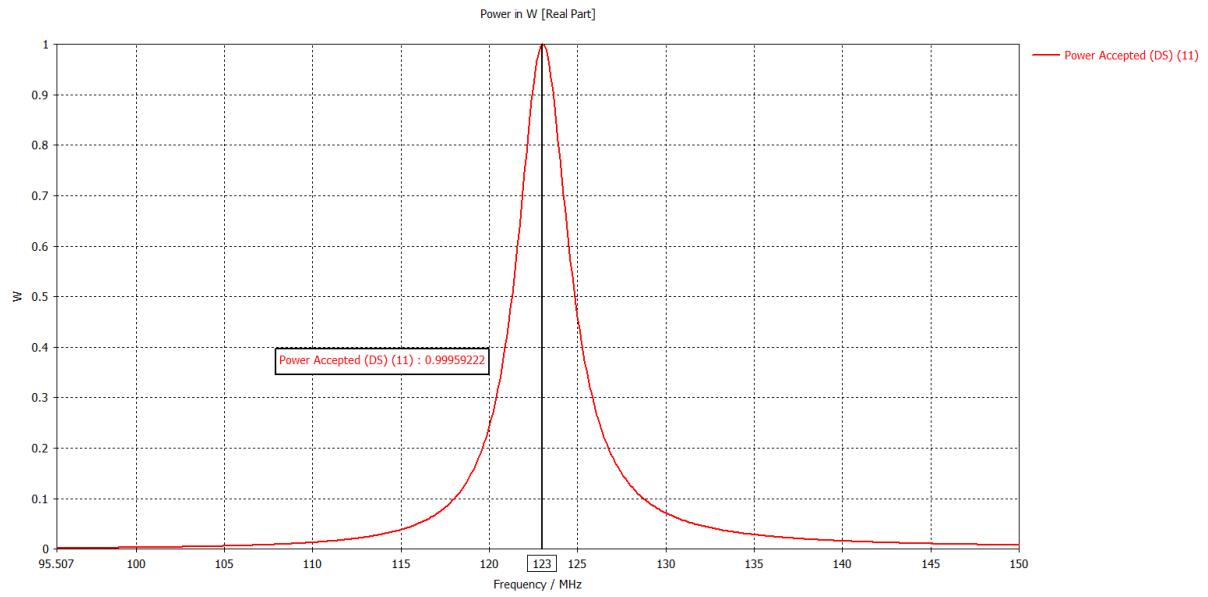


Figure 25: Frequency dependence of the accepted power. The curve peaks at 123 MHz with a value of 0.99959222, confirming efficient power delivery at the Larmor frequency.

E Outgoing Power – 1D Plot

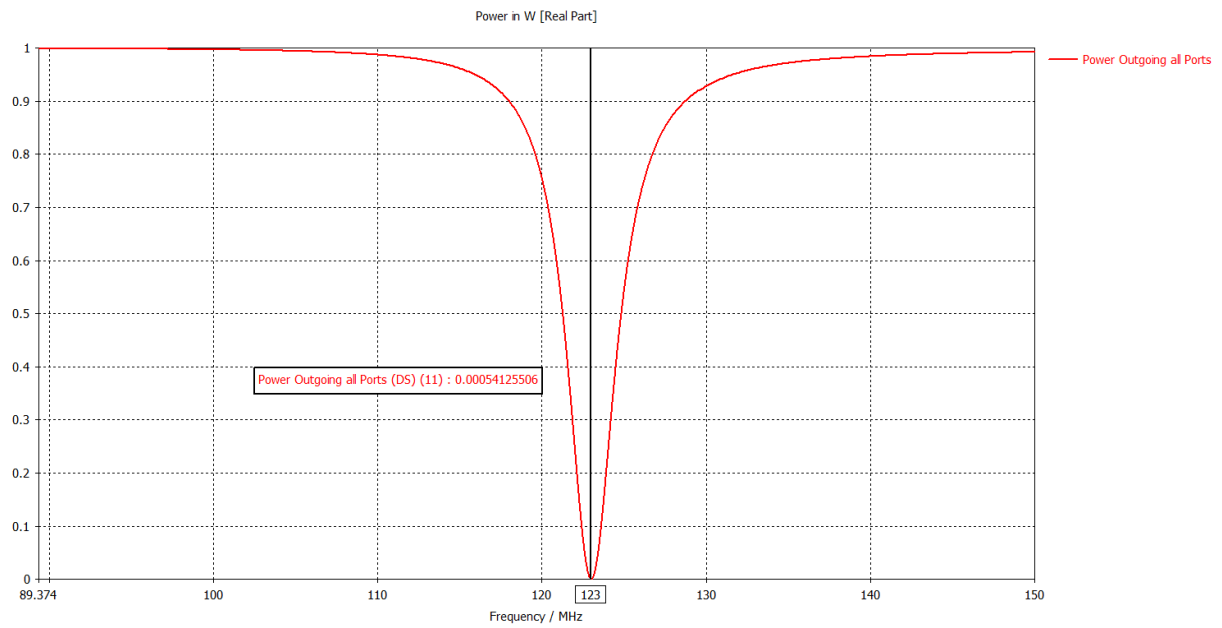


Figure 26: Frequency dependence of the total outgoing power from all ports. The curve shows a minimum at 123 MHz with a value of 0.00054125506, indicating very low reflection and port leakage at the operating frequency.